Statement of Work

Project: Integrated Crime & Social Analytics Pipeline (2019-2024)
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Goals and Objectives

The primary goal of this project was to prepare high-quality data that could be ingested into a Business Intelligence (BI) tool to support monitoring, comparison, and prioritisation across police forces using a scalable and robust data handling pipeline

With the primary goal in mind, the following objectives were set to achieve this:

- Ingest and clean longitudinal crime data
- Integrate enrichment data for wider force comparison
- Calculate per-capita crime rates
- Aggregate crime records to a Police Force X Year X x LSOA Name X Crime grain

Project Scope

Target Forces: Metropolitan, Surrey, Sussex and Thames Valley Police

Geographic Value: These forces surround the London commuter belt, enabling analysis of how crime spills over from a major metropolitan hub into neighbouring county footprints.

Size Contrast: The selection puts a vast force (Metropolitan) against smaller regional sectors, creating the perfect test case for per-capita rate normalisations.

Temporal Window: Spans a strict six-year longitudinal window from January 2019 through December 2024 to evaluate shifting crime patterns across differing stages of the COVID-19 pandemic

Data Sources

Dataset	Type	Source	Analytical Purpose	Structure
UK Police Crime Data	Primary	UK Police Data Portal	Details exactly what crimes are happening, their total volumes, as well as their shift over time	Consists of a folder for every month of the year (2019-2024) containing csv files for each force's crime data for that month
Population Data	Enrichment	Office For National Statistics	Allows for normalisation of the data into crimes per 1000 residents, so	Consists of multiple sheets which each contain population data between a certain range

			different-sized forces can be compared head-to-head fairly	
Deprivation or socioeconomic indicators	Enrichment	English Indices of Deprivation 2025	Identifies if local poverty, employment, or housing issues directly influence	Consists of a single table

Technical Methodology

The pipeline used to transform and aggregate all of the raw data followed this pipeline:

Data Cleaning

The crime data is pulled via an iterative file-system batch processing loop to protect system memory, dynamically appending parent directory strings as the 'Month' attribute. drop_duplicates() was applied to filter out 1,011,718 redundant rows. Columns with null values were handled by applying placeholder values to retain the existing data and avoid skewing future analysis if the data were dropped instead. The 'Context' and 'Falls within' columns were dropped since these were empty and fully duplicated, respectively. All text columns were stripped of trailing whitespaces and capitalised uniformly to ensure consistency between all the values.

Feature Engineering & Enrichment Layer

A numeric 'Year' column was derived via pd.to_datetime(). The raw population data tallied the population by individual age and gender, so a .sum(axis=1) operation was performed to tally the total population for a force for a given year as 'Population'. In the raw data deprivation data, the 'is_decile_1' flag was created to flag if a neighbourhood was one of the top 10% most deprived areas nationally. The data was combined using pd.merge() left joins on explicit keys (['Force', 'Year')] for the population data and ['Force'] for the deprivation data to ensure that no "many-to-many" connections occurred.

Data Aggregation

The enriched primary dataset is compressed using a .groupby() method to collapse the individual records into a table representing the grain defined in the scope: Force X Year X LSOA Name X Crime type. Volumes were summarised using count(), whilst latitude and longitude were aggregated using mean() to obtain an average coordinate for crime rather than unique coordinates, which would explode the number of rows. Regional per-capita normalisations are evaluated using the formula:
Force Crimes per 1,000 residents = (Total Annual Force Crime Count / Annual Force Population Estimate) x 1000

Constraints and Assumptions

Crimes missing LDOA (neighbourhood) assignments were allocated placeholder strings. It is assumed that downstream report developers will filter these out when conducting geographic analysis using LDOA

While crime data is logged monthly, ONS population counts are published annually. The pipeline assumes force-level residential populations remain static month to month across the calendar year to serve as a stable baseline